A Study in Banking Competition: Illinois banking market between 1984 and 2020

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Abstract

The objective of the following study is to provide an assessment on the degree of competition in the banking sector in the American state of Illinois between 1984 and 2020[1]. This study utilizes widely-adopted models and contemporary estimation strategies to generate significant empirical evidence that can be generalized to different scenarios. Differently from the existing literature, this empirical analysis addresses issues related to geographical and time heterogeneity and returns economically meaningful results that show the evolution of competition and provide insights into the differences emerging from geographical distributions.

I. Introduction

For most of the 20th century, Illinois was strictly a unit banking state. Due to state laws that prohibited the opening of branches over the whole territory, the banking sector has developed in a largely fragmented and specialized way. The persistence of the effects of such legislation can still be widely observed today many years after its abolition in 1993[2]. The prevailing model for banking entities located outside Illinois' primary cities is the independent unit bank, which operates as a singular local presence and extends its services to a substantial rural territory.

This first historical fact already underlines the need for a cautious methodological approach when studying the banking sector in the mid-west state. By taking into account both the temporal and the geographical dimension, the paper is able to distinguish between collusive or competitive behaviors in large cities against their rural counterparts and the evolution of such behaviors in different epochs. Cross comparison of these findings can prove the consequences of long-lasting inefficient systems and provide insights into the geographical patterns that competition follows. Relying on the assumptions laid by Stigler (1964) and later adopted by Bikker and Bos (2005) and Bos et al. (2017), I empirically show the widespread collusive behavior adopted by Illinois banks, its positive correlation with spatial isolation (Salop, 1979) and its reaction to changes in legislation.

[1] The data used for the purpose of this paper is a collection of call reports quarterly published by Illinois' banks and subsequently collected from Wharton Research Data Services (WRDS).

[2] Already the Public Act 85-204 (1987) repealed the Section 6 of Illinois Banking Act explicitly against branching. However, the limitation on the numbers and location of branches still in place was officially removed in 1993 with the passing of the Public Act 88-4.

II. Empirical Approach

The largest theoretical foundation used for this empirical research is provided by the Critical Mass Model (Bos et al., 2017) built upon the theoretical work of Cowling and Waterson (1976) and Cowling (1976). Avoiding the assumption of homogeneous conjectural variation, and without using the Hirschman-Herfindahl Index as a proxy for the banks' market share, I find these models as the most appropriate and precise for the purpose of my empirical research. As they do not assume homogeneity over time and between cities, these models are exceptional at providing entity singular estimates of: conjectural variation (λ_{it}), market share (θ_{it}) and profit extraction (L_{it}) for each time period. The Cowling and Waterson (1976) computation of mark-up Lerner index is obtained from a combination of the mentioned variables and of the inverse of the elasticity of demand a bank faces:

$$L_{it} = \left(-\frac{1}{\eta}\right)\theta_{it}(1+\lambda_{it}) \tag{1}$$

The Bos et al. (2017) adaptation for empirical research:

$$L_{it} = \beta_i + \beta_\theta \theta_{it} + \beta_\lambda \lambda_{it} + \beta_{\theta\cdot\lambda}(\theta_{it}\cdot\lambda_{it}) + \beta_x \text{Controls} + \epsilon_{it} \tag{2}$$

and the critical mass θ^* calculated as:

$$\theta^* = -\frac{\beta_\lambda}{\beta_{\theta\cdot\lambda}} \tag{3}$$

This identified threshold is defined as the point in which the penalty of cheating exceeds the price cutting (Stigler, 1964), i.e. the point in which it becomes more convenient for a bank to collude rather than increase its customer base through decreases in price. The penalty is defined as the losses accrued if such decreases in price trigger a price war.

Identifying the geographic discontinuities between metropolitan cities (MSA) and rural areas, the time periods over which I subdivide the study are four. The first one is between the beginning of the sample, 1984, and the abolition of the anti-branching legislation in 1993 (moment in which both HHI and the shares of the top 5 banks are at their minimum). The subsequent period (1994-2006) is historically characterized by large mergers in the banking sector and observed to culminate in a spiking concentration point (Figure 1). The financial crisis (2007-2010) is treated separately and the post-crisis high regulatory period (2010-2020) is the last analyzed time period of the study.

The empirical strategy essentially consists in the repetition of the Bos et al. (2017) critical mass exercise over these four periods assuming geographical homogeneity, on the two distinguished areas assuming time homogeneity and the comparison of these with the baseline state specification.[3] Elasticity of demand (η) is assumed to be constant throughout and estimated from the changes in quantities as responses to changes in prices. It is found to be ≈ -0.454[4]:

$$\log(Q_{it}) = \alpha_i + \theta_t + \beta\log(P_{it}) + \gamma\text{Controls}_{it} + \epsilon_{it} \tag{4}$$

[3] In both cases regressions are estimated using time and entity fixed effects. Although these within-entity and across-time trends are accounted for, the division in multiple regression allows to have different slopes (and thus different θ^*'s) for each period and area, rather than yielding simple intercepts.

[4] As in Shaffer (1989) elasticity of demand is estimated using a 2SLS regression using cost of funds (interest rates paid on deposits) as instrument.

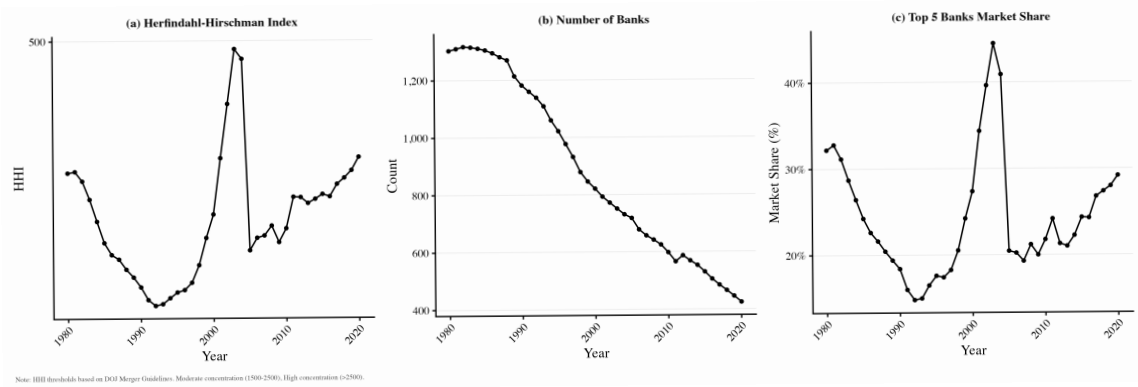

Figure 1: Illinois's Banking System Evolution

An accurate calculation of marginal costs (MC) involves the need of highly precise variables not at public disposition, or the estimation of a complex traslog function that often suffers from multiple biases. Because true economic marginal cost is unobservable in standard financial reporting, I follow the established empirical literature (Domowitz et al., 1986; Hall, 1988) by utilizing Average Variable Cost (AVC) as a robust accounting proxy. Therefore, Marginal Cost is approximated as:

$$MC_{it} \approx AVC_{it} = \frac{\text{Total Interest Expenses}_{it}}{\text{Total Deposits}_{it}} + \frac{\text{Total Non Interest Expenses}_{it}}{\text{Total Assets}_{it}} \tag{5}$$

From which I can extrapolate the Lerner mark-up indexes from each bank in each time period:

$$L_{it} = \frac{P_{it} - MC_{it}}{P_{it}} \tag{6}$$

The Lerner indexes are then used to approximate the conduct parameter, or conjectural variation (λ_{it}), using the Cowling and Waterson (1976) equation (1). The definition of these variables allows for the implementation of the empirical regression which, however, must be corrected for the time period. Extrapolating all these values simultaneously and forcing deterministic relationships breaks the statistical algorithm which therefore finds no variation at all. By lagging the market share ($\theta_{i,t-1}$) and the conjectural variation ($\lambda_{i,t-1}$) by one quarter I am able to break this algebraic trap and gain a deeper economic meaning: the market share and the conduct that a bank had in the previous period dictates the setting of prices and thus the mark-up in this period.

III. Results

As visible from Table 1, the near-zero critical mass (θ^*) found by the estimation is exceeded by all banks, across all periods and in every geographical area. This result confirms the initial hypothesis by showing that state-wide banking competition parameters are not accurate metrics to observe rent extraction behaviors within the state of Illinois. The persistent effects of the unitary banking tradition provide evidence in favor of the usage of city specific variables instead of state-wide ones. As shown in Panel A, comparing small local banks able to exert some degree of

Table 1: **Market Power Metrics Across Sub-Samples**
Temporal and Geographical Heterogeneity in Illinois Banking Competition

Sub-Sample	N (Obs)	N (Banks)	Critical Mass (θ^*)[1]	% Banks with Market Power[2]	Avg # Banks with Power[3]	Marginal Effect[4]
Panel A: Geographical Analysis						
Rural	74,343	1,136	0.000002	100.00	30.70	771.49
Urban	38,841	876	0.000004	100.00	43.80	333.93
Panel B: Temporal Analysis						
1984-1993	44,189	1,325	0.000006	100.00	132.50	459.67
1994-2006	40,327	1,117	0.000001	100.00	85.92	775.01
2007-2010	9,481	636	0.000000	100.00	159.00	373.43
2011-2020	19,187	592	0.000001	100.00	59.20	760.42
Baseline						
Full Sample	113,184	1,525	0.000002	100.00	41.22	619.97

1. Critical mass market share where collusion becomes profitable
2. Percentage of observations with Lag Market Share $\geq \theta^*$
3. Average number of unique banks exceeding θ^* per year
4. Marginal effect of market share on Lerner Index at mean conduct parameter

spatial monopoly power with the metropolitan international banks can produce misleading results. Even if all banks in both samples are perceived to have market power, the critical mass for banks outside the largest metropolitan areas is half of the one required to have market power within cities. Moreover, banks located in rural contexts are observed to be able to extract double the amount of rents for each additional point of market share acquired compared to the urban counterparts. These findings point out the existence of spatially driven monopolistic competition (Salop, 1979) by showing that spread in distances between two banks facilities is associated with higher consumer surplus extraction (see Appendix Figure 1). The model, however, cannot account for the impact of distance on unmeasurable characteristics such as information gathering or monitoring costs. These factors might positively affect the interest rate charged on borrowers to balance higher risks or higher costs rather than reflecting strict monopoly prices (Petersen and Rajan, 1995). Panel B presenting the temporal analysis yields interesting results as well. In the banking competition theoretical framework there are two major theories associating number of banks with degree of competition and rent extraction. Following Allen and Gale (2003) I expected that a reduction in the number of banks should have been associated with smaller possibilities of rent extraction at each market share level thanks to the benefits consumers gain with branching (allowed after 1993). The standard Cournot model predicts instead that reducing the number of banks operating in a market push them to adopt an oligopolistic behavior more prone to extract rents. The fluctuating behavior of the marginal effect (Column 6) against a constant decrease in the number of banks (Column 2) seems to invalidate both theories and to point out the existence of other driving forces (see Appendix Figure 2). The decrease in banking competition emerging after 1993 may also be a direct consequence of the resolution of the Savings & Loan crisis. As government provided liquidation of insolvent thrifts, the destructive competition for deposits previously caused by failing institutions was eliminated. This allowed surviving commercial banks to restore their margins and increase

their rent extraction capabilities (Berger et al., 1995). This same trend could have been boosted by the Riegle-Neal Act (1994): allowing for interstate banking may have increased competition only in metropolitan areas while helping rural banks to consolidate further (Critchfield et al., 2004). As for the first analyzed period, the decrease in profits experienced between 2007 and 2010 are symptoms of economic crisis and competitive pressure. The increasing strictness of banking regulation emerging afterwards can be a factor influencing banks' loan availability, fixed costs and entry barriers (Adams and Gramlich, 2014), thus lowering competition and enabling healthy banks to shift additional costs onto their customers.

IV. Conclusion

Employing rigorous and tailored econometric techniques, the analysis highlighted a significant aspect of Illinois's banking infrastructure. The minimal degree of competition found in all seven specifications carries relevant insights for policy debate and future regulation decisions. The persistence of inefficiencies caused by unitary banking system that enabled the retention of monopolistic power by banks established outside of the largest cities is certainly the most remarkable result. The abolition of the McFadden Act in 1994 had observable effects in the highly populated areas and in the Chicago's financial hub but did not carry any revolution in the provincial surroundings. This evidence is furthermore supported by an additional analysis aimed at disentangling the evolution of area-specific outcomes within each time period. The marginal effect of the market share on Lerner index is found at its highest level (1,013) between 1994 and 2006 in rural areas, compared its counterpart in urban areas estimated at a level of 372. Differences of this kind underscore the need of using geographical characteristics as central determinants in targeting and assessing the impact of legislation in the banking sector.

While additional research on Illinois's banking sector could provide valuable insights and potential diverging results by finding better approximation of marginal costs, differences in demand elasticity across time and space, or an annual evolution of the variables of interest; this analysis has shown the evolution of the competition in the mid-west banking sector highlighting its high degree of monopolistic competition. Evaluation of the trade-offs between widespread market power potentially contributing to financial stability and high competition levels protecting consumers' interests is left for policymakers.

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Appendix

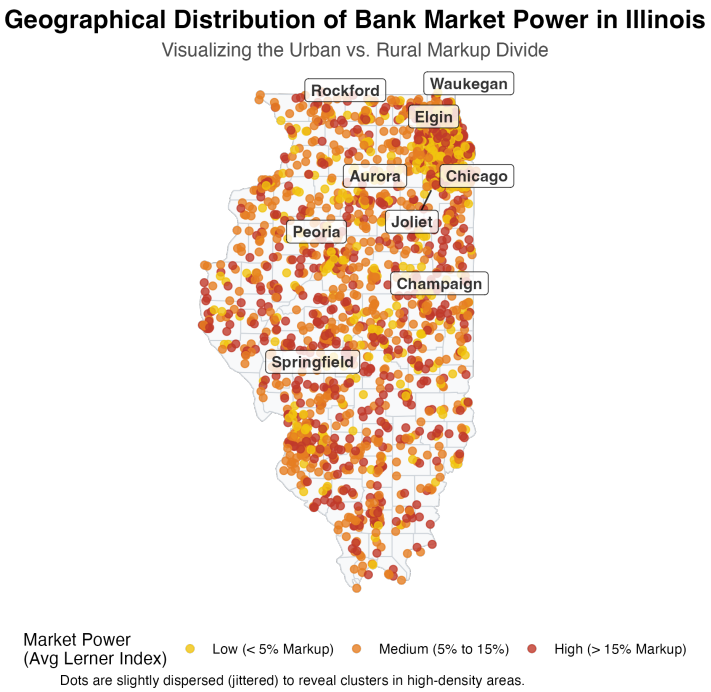

Figure 1: The image highlights the negative relation existing between markup intensity and large city proximity.

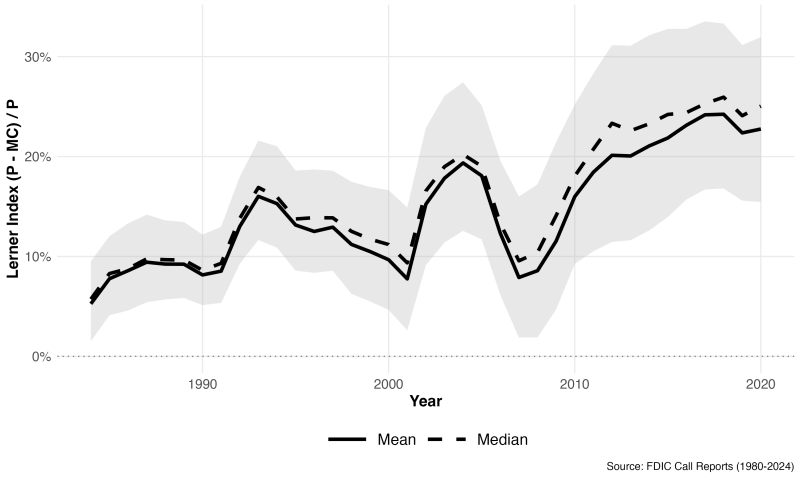

Figure 2: Lerner Index evolution over time suggests the absence of any correlation between this variable and the number of banks operating in Illinois.